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Battery rack having fixing frame and energy storage system

Abstract

A battery rack with increased durability to prevent a rack case from being deformed or broken by external impacts during delivery. To achieve the above-described object, the battery rack includes a plurality of battery modules arranged in a first direction, at least one rack case including a plurality of rack frames configured to mount the plurality of battery modules, and at least one fixing frame including a body in the shape of a plate extending along the plurality of battery modules and a fixing portion formed on one surface of the body and configured to be attached and detached to/from each of the plurality of battery modules.

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Background/Summary

TECHNICAL FIELD
(1) The present disclosure relates to a battery rack having a fixing frame, and more particularly, to a battery rack with increased durability to prevent a rack case from being deformed or broken by external impacts during delivery.
(2) The present application is the National Phase Application of PCT/KR2020/095096, filed on Jul. 27, 2020, and claims the benefit of Korean Patent Application No. 10-2019-0105127 filed on Aug. 27, 2019 and Korean Patent Application No. 10-2020-0090588 filed on Jul. 21, 2020 with the Korean Intellectual Property Office, the disclosures of which are incorporated herein by reference in its entirety.
BACKGROUND ART
(3) Currently, commercially available secondary batteries include nickel-cadmium batteries, nickel-hydrogen batteries, nickel-zinc batteries, lithium secondary batteries and the like, and among them, lithium secondary batteries have little or no memory effect, and thus they are gaining more attention than nickel-based secondary batteries for their advantages that recharging can be done whenever it is convenient, the self-discharge rate is very low and the energy density is high.
(4) A lithium secondary battery uses lithium-based oxide and a carbon material for a positive electrode active material and a negative electrode active material respectively. The lithium secondary battery includes an electrode assembly including a positive electrode plate and a negative electrode plate respectively coated with the positive electrode active material and the negative electrode active material with a separator interposed between, and a packaging or a battery pouch case to hermetically receive the electrode assembly together with an electrolyte solution.
(5) Recently, secondary batteries are widely used not only in small devices such as portable

electronic devices, but also in medium- and large-sized devices such as vehicles and energy storage systems. For use in medium- and large-sized device applications, many secondary batteries are electrically connected to increase the capacity and output. In particular, pouch-type secondary batteries are widely used in medium- and large-sized devices due to their advantage of being easy to stack.

(6) More recently, with the growing need for large-capacity structures for use as a source of energy, there is an increasing demand for a battery rack including a plurality of secondary batteries electrically connected in series and/or in parallel, a battery module in which the secondary batteries are received and a battery management system (BMS).

(7) The battery rack generally includes a rack case of metal to protect the battery modules from external impacts or receive and store them. Moreover, in recent years, with the growing demand for high capacity battery racks, there is an increasing trend towards battery racks including a plurality of battery modules having heavy loads.

(8) However, the battery rack including the plurality of heavy battery modules has a very great weight. By this reason, during delivery of the battery rack including the plurality of battery modules received in the rack case, the rack case is deformed or broken by the loads of the heavy battery modules when the rack case vibrates or tilts.

(9) To prevent the rack case from being deformed during delivery of the battery rack, the plurality of battery modules and the rack case are separately delivered. However, in this case, they occupy a higher volume, which increases the delivery cost, and it is necessary to assemble the rack case and the battery module in the installation location. As a consequence, the production cost of the battery rack increases.

DISCLOSURE

Technical Problem

(10) The present disclosure is designed to solve the above-described problem, and therefore the present disclosure is directed to providing a battery rack with increased durability to prevent a rack case from being deformed or broken by external impacts during delivery.

(11) These and other objects and advantages of the present disclosure can be understood by the following description, and will be apparent from the embodiments of the present disclosure. In addition, it will be readily appreciated that the objects and advantages of the present disclosure can be realized by means and combinations thereof.

Technical Solution

(12) To achieve the above-described object, a battery rack according to the present disclosure includes a plurality of battery modules arranged in a first direction, at least one rack case including a plurality of rack frames configured to mount the plurality of battery modules, and at least one fixing frame including a body in the shape of a plate extending along the plurality of battery modules and a fixing portion formed on one surface of the body and configured to be attached to each of the plurality of battery modules or detached from each of the plurality of battery modules.

(13) Additionally, the battery module may include a plurality of hanger members respectively coupled to an outer surface of the plurality of battery modules, the fixing portion of the fixing frame may have a plurality of hook structures configured to be hook-coupled to the plurality of hanger members, and the plurality of hanger members may be configured to be coupled to a respective one of the plurality of hook structure of the fixing portion of the fixing frame.

(14) Additionally, each of the plurality of battery module may include a module housing including an upper cover, a base plate, a front cover and a rear cover, and the each hanger member of the plurality of hanger members may be disposed on an outer side of the rear cover of a respective one of the plurality of module housings.

(15) Additionally, each hanger member of the plurality of hanger members may include a hanger bar in the shape of a plate in which that the hook structure is inserted and fixed on one side, the hanger bar spaced a predetermined distance apart from the outer surface of the battery module, and

a leg extending from each of two ends of the hanger bar, wherein each leg is fixed to the outer surface of the battery module, and each leg has a bent portion.
(16) Additionally, the at least one fixing frame may include a support portion extending vertically along the body and extending from the body such that a surface wraps around at least part of an outer surface of each leg.
(17) Additionally, the support portion may be configured to wrap around at least part of the outer surface of each leg of the hanger member even when the hook structure is separated from the hanger bar of the hanger member.
(18) Additionally, upper and lower ends of the body of the at least one fixing frame may have a case coupling portion that is coupled to the rack case.
(19) Additionally, the rack case may include a connecting bar connected to the case coupling portion to couple the case coupling portion to the rack case.
(20) Additionally, the case coupling portion of the at least one fixing frame may include an upper bolting hole configured to couple the plurality of hook structures to the plurality of hanger members when coupled to the connecting bar, and a lower bolting hole spaced from the upper bolting hole at a predetermined distance in the vertical direction and configured to separate the plurality of structures from the plurality of hanger members when coupled to the connecting bar.
(21) Additionally, the fixing frame may further include an indicator configured to indicate whether the plurality of hook structure structures is coupled to the plurality of hanger members.
(22) Additionally, the fixing frame may further include a wall coupling portion configured to fix the fixing frame to a wall adjacent to the battery rack.
(23) Additionally, the rack case may include a front frame including a plurality of pillars disposed at a front end of the plurality of battery modules and extending in a vertical direction and a connecting part connecting the pillars in a horizontal direction, a rear frame including a plurality of pillars disposed at a rear end of the plurality of battery modules and extending in the vertical direction and a connecting part connecting the pillars in the horizontal direction, and a top plate disposed on top of the plurality of battery modules.
(24) Additionally, a lower end of each of the front frame and the rear frame may have a bent portion having one surface parallel to the ground, and a bolting hole formed in the bent portion.
(25) Additionally, the battery rack may include two or more rack cases stacked in a vertical direction, each pillar of the front frame and the rear frame may have a bolting hole configured to be coupled to another rack case at an upper end, and a bolting hole may be disposed at the bent portion of the pillar the rack case disposed at an upper position may be bolt-coupled to a bolting hole of the pillar of the rack case disposed at a lower position.
(26) Additionally, each pillar may include an outer pillar extending in a vertical direction and having one open horizontal side and an inner pillar inserted into the outer pillar and extending in the vertical direction.
(27) Additionally, the outer pillar may be a C-shaped steel that is open to one side when viewed in a horizontal cross section, and the inner pillar may a rectangular pipe.
(28) Additionally, the bent portion of the rear frame may have a slit into which a body of a bolt is slidably inserted in a horizontal direction.
(29) To achieve the above-described object, an energy storage system according to the present disclosure includes at least one battery rack.
Advantageous Effects
(30) According to an aspect of the present disclosure, the battery rack of the present disclosure includes at least one fixing frame including the fixing portion configured to be attached/detached to/from each of the plurality of battery modules, to prevent deformation of the rack case or failure or damage of the battery module or separation of the battery module from the battery rack due to the movement by the heavy battery module during delivery of the battery rack.
(31) According to an aspect of an embodiment of the present disclosure, the present disclosure is

configured such that the hanger member installed in each of the plurality of battery modules and the fixing portion of the fixing frame are hook-coupled to each other, it is possible to easily couple to each of the plurality of battery modules when the fixing frame hangs on the hanger member. When the delivery of the battery rack is completed and it is necessary to install in the installation location, it is possible to easily decouple the fixing frame from each of the plurality of battery modules, thereby performing the separation operation in a simple and quick manner.
(32) According to an aspect of the present disclosure, the fixing frame of the present disclosure further includes the support portion extending from the body such that one surface wraps around at least part of the outer surface of the leg, to prevent the battery module from moving to the left and right during delivery. Accordingly, it is possible to prevent deformation of the rack case caused by the leftward and rightward movement of the battery module.
(33) According to another aspect of the present disclosure, the wall coupling portion of the fixing frame is configured to fix the fixing frame to the wall adjacent to the battery rack, to couple and fix the upper part of the rack case to the wall of the storage location. Accordingly, it is possible to prevent the upper part of the battery rack from moving back and forth and to the left and right during delivery. Accordingly, it is possible to prevent deformation of the rack case of the battery rack, or deformation or damage of the plurality of battery modules by the rack case.
(34) According to an aspect of the present disclosure, the pillar includes the outer pillar extending in the vertical direction and having one open horizontal side, and the inner pillar inserted into the outer pillar and extending in the vertical direction, thereby effectively increasing the mechanical strength of the rack case. That is, the combination of the inner pillar and the outer pillar, not a single pillar, may ensure a higher mechanical strength than the single configuration without increasing the weight of the rack case.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The accompanying drawings illustrate preferred embodiments of the present disclosure and, together with the foregoing disclosure, serve to provide further understanding of the technical spirit of the present disclosure. However, the present disclosure is not to be construed as being limited to the drawings.
(2) FIG. **1** is a schematic front perspective view of a battery rack according to an embodiment of the present disclosure.
(3) FIG. **2** is a schematic rear perspective view of a battery rack according to an embodiment of the present disclosure.
(4) FIG. **3** is a schematic exploded perspective view of a battery rack according to an embodiment of the present disclosure.
(5) FIG. **4** is a schematic exploded perspective view of the inner part of a battery module of a battery rack according to an embodiment of the present disclosure.
(6) FIG. **5** is a schematic rear perspective view of a battery module of a battery rack according to an embodiment of the present disclosure.
(7) FIG. **6** is a schematic rear perspective view of a fixing frame of a battery rack according to an embodiment of the present disclosure.
(8) FIG. **7** is a schematic front perspective view of a fixing frame of a battery rack according to an embodiment of the present disclosure.
(9) FIG. **8** is a schematic partial plane view of a battery module and a fixing frame according to an embodiment of the present disclosure.
(10) FIG. **9** is a schematic enlarged plane view of section A in FIG. **8**.
(11) FIG. **10** is a schematic rear perspective view of a fixing frame according to another

embodiment of the present disclosure.

(12) FIG. **11** is a schematic partial perspective view of a lower part of a battery rack according to an embodiment of the present disclosure.

(13) FIG. **12** is a schematic partial perspective view of an upper part of a battery rack according to an embodiment of the present disclosure.

(14) FIG. **13** is a schematic partial perspective view of a battery rack according to another embodiment of the present disclosure.

(15) FIG. **14** is a schematic partial horizontal cross-sectional view of the battery rack of FIG. **1** taken along the line A-A′.

(16) FIGS. **15** and **16** are schematic rear views of a fixing frame of a battery rack according to another embodiment of the present disclosure.

DETAILED DESCRIPTION

(17) Hereinafter, the preferred embodiments of the present disclosure will be described in detail with reference to the accompanying drawings. Prior to the description, it should be understood that the terms or words used in the specification and the appended claims should not be construed as being limited to general and dictionary meanings, but rather interpreted based on the meanings and concepts corresponding to the technical aspects of the present disclosure on the basis of the principle that the inventor is allowed to define the terms appropriately for the best explanation.

(18) Therefore, the embodiments described herein and illustrations shown in the drawings are just a most preferred embodiment of the present disclosure, but not intended to fully describe the technical aspects of the present disclosure, so it should be understood that a variety of other equivalents and modifications could have been made thereto at the time that the application was filed.

(19) FIG. **1** is a schematic front perspective view of a battery rack according to an embodiment of the present disclosure. FIG. **2** is a schematic rear perspective view of the battery rack according to an embodiment of the present disclosure. FIG. **3** is a schematic exploded perspective view of the battery rack according to an embodiment of the present disclosure. FIG. **4** is a schematic exploded perspective view of the inner part of a battery module of the battery rack according to an embodiment of the present disclosure.

(20) Referring to FIGS. **1** to **4**, the battery rack **300** according to an embodiment of the present disclosure includes a plurality of battery modules **200** arranged in a direction, at least one rack case **310** and at least one fixing frame **320**.

(21) Here, the plurality of battery modules **200** may be received within the rack case **310** in the vertical arrangement.

(22) The plurality of battery modules **200** may include a module housing **210**, and a cell assembly **100** having a plurality of secondary batteries **110** provided in the module housing **210** and stacked in a direction.

(23) In detail, the secondary battery **110** may be a pouch-type secondary battery. For example, as shown in FIG. **4**, each of 2 cell assemblies **100** may include 21 pouch-type secondary batteries **110** stacked side by side in the front-rear direction (y direction).

(24) In particular, the pouch-type secondary battery **110** may include an electrode assembly (not shown), an electrolyte solution (not shown) and a pouch.

(25) Each secondary battery **110** may stand in a direction (z direction) approximately perpendicular to the ground with two wide surfaces disposed in the front and rear directions and sealing portions disposed in the up, down, left and right directions.

(26) Here, the pouch may have a concave receiving portion. The electrode assembly and the electrolyte solution may be received in the receiving portion. Each pouch may include an outer insulating layer, a metal layer and an inner insulating layer, and the inner adhesive layers adhere to each other at the edges of the pouch to form a sealing portion. A terrace portion may be formed at each end in the left-right direction (x direction) where a positive electrode lead **111** and a negative

electrode lead (not shown) of the secondary battery **110** are formed.
(27) The electrode assembly is an assembly of an electrode plate coated with an electrode active material and a separator, and includes at least one positive electrode plate and at least one negative electrode plate with a separator interposed between. The positive electrode plate of the electrode assembly may have a positive electrode tab, and at least one positive electrode tab may be connected to the positive electrode lead **111**.
(28) Here, the positive electrode lead **111** may have one end connected to the positive electrode tab and the other end exposed to the outside through the pouch, and the exposed portion may act as an electrode terminal of the secondary battery **110**, for example, a positive electrode terminal of the secondary battery **110**.
(29) The negative electrode of the electrode assembly may have a negative electrode tab, and at least one negative electrode tab may be connected to the negative electrode lead. The negative electrode lead may have one end connected to the negative electrode tab and the other end exposed to the outside through the pouch, and the exposed portion may act as an electrode terminal of the secondary battery **110**, for example, a negative electrode terminal of the secondary battery **110**.
(30) As shown in FIG. **4**, the positive electrode lead **111** and the negative electrode lead may be formed at the left and right ends (x direction) with respect to the center of the secondary battery **110**. That is, the positive electrode lead **111** may be provided at one end (right end) with respect to the center of the secondary battery **110**. The negative electrode lead may be provided at the other end (left end) with respect to the center of the secondary battery **110**.
(31) For example, as shown in FIG. **4**, each secondary battery **110** of the cell assembly **100** may be configured such that the positive electrode lead **111** and the negative electrode lead extend in the left-right direction.
(32) According to this configuration of the present disclosure, it is possible to increase the area of the electrode lead without interference between the positive electrode lead **111** and the negative electrode lead of one secondary battery **110**.
(33) The positive electrode lead **111** and the negative electrode lead may be formed in the shape of a plate. In particular, the positive electrode lead **111** and the negative electrode lead may extend in the horizontal direction (X direction) with the wide surfaces standing upright in the front-rear direction.
(34) Here, the horizontal direction refers to a direction parallel to the ground when the battery module **200** is placed on the ground, and may be referred to as at least one direction on a plane perpendicular to the vertical direction.
(35) However, the battery module **200** according to the present disclosure is not limited to the above-described pouch-type secondary battery **110** and may include a variety of secondary batteries **110** known at the time of filing the patent application.
(36) The at least two cell assemblies **100** may be arranged in the front-rear direction. For example, as shown in FIG. **4**, 2 cell assemblies **100** may be arranged in the front-rear direction, and may be spaced a predetermined distance apart from each other.
(37) The battery module **200** may further include a busbar assembly **280**. In detail, the busbar assembly **280** may include at least one busbar **282** configured to electrically connect the plurality of secondary batteries **110** and at least two busbar frames **286** configured to mount the at least one busbar **282** on the outside. The at least two busbar frames **286** may be respectively provided on the left and right sides of the cell assembly **100**.
(38) In detail, the busbar **282** may include a conductive metal, for example, copper, aluminum and nickel.
(39) The busbar frame **286** may include an electrical insulating material. For example, the busbar frame **286** may include a plastic material. In more detail, the plastic material may be polyvinyl chloride.
(40) The module housing **210** may have an internal space to receive the cell assembly **100** therein.

In detail, the module housing **210** may include an upper cover **220**, a base plate **240**, a front cover **250** and a rear cover **260**.
(41) In detail, the base plate **240** may have a larger area than the size of the lower surface of the at least two cell assemblies **100** to mount the at least two cell assemblies **100** thereon. The base plate **240** may be in the shape of a plate extending in the horizontal direction.
(42) The upper cover **220** may include a top **224** and a side **226**. The top **224** may be in the shape of a plate extending in the horizontal direction to cover the top of the cell assembly **100**. The side **226** may be in the shape of a plate extending in the downward direction from the left and right ends of the top **224** to cover the left and right sides of the cell assembly **100**.
(43) The side **226** may be coupled to a portion of the base plate **240**. For example, as shown in FIG. **3**, the upper cover **220** may have the top **224** in the shape of a plate extending in the front-rear direction and the left-right direction. The upper cover **220** may have 2 sides **226** extending in the downward direction from each of the left and right ends of the top **224**. The lower end of each of the 2 sides **226** may be configured to be coupled to the left and right ends of the base plate **240**. In this instance, the coupling method may be a male-female coupling method or a welding coupling method.
(44) The front cover **250** may be configured to cover the front side of the plurality of secondary batteries **110**. For example, the front cover **250** may be in the shape of a plate having a larger size than the front surface of the plurality of secondary batteries **110**. The plate shape may stand in the vertical direction.
(45) Part of the outer periphery of the front cover **250** may be coupled to the base plate **240**. For example, the lower part of the outer periphery of the front cover **250** may be coupled to the front end of the base plate **240**. The upper part of the outer periphery of the front cover **250** may be coupled to the front end of the upper cover **220**. Here, the coupling method may include bolt coupling.
(46) The rear cover **260** may be configured to cover the rear side of the cell assembly **100**. For example, the rear cover **260** may be in the shape of a plate having a larger size than the rear surface of the plurality of secondary batteries **110**.
(47) Part of the outer periphery of the rear cover **260** may be coupled to the base plate **240**. For example, the lower part of the outer periphery of the rear cover **260** may be coupled to the front end of the base plate **240**. The upper part of the outer periphery of the rear cover **260** may be coupled to the rear end of the upper cover **220**. Here, the coupling method may include bolt coupling.
(48) According to this configuration of the present disclosure, the module housing **210** is configured to stably protect the plurality of secondary batteries **110** from external impacts, to increase the safety of the battery module **200** against external impacts.
(49) Referring back to FIGS. **1** and **2**, the at least one rack case **310** may include a plurality of rack frames **315** configured to mount the plurality of battery modules **200**, a front frame **311** disposed at the front end of the plurality of battery modules **200**, and a rear frame **313** disposed at the rear end of the plurality of battery modules **200**.
(50) In detail, the rack frame **315** includes a receiving plate **315***a*. The receiving plate **315***a* may be in the shape of a plate that is bent about 90° in the shape of ‘L’. The front and rear ends of the receiving plate **315***a* may be configured to be connected to the front frame **311** and the rear frame **313** respectively.
(51) For example, as shown in FIG. **2**, 2 receiving plates **315***a* may be provided to receive one battery module **200**. The 2 receiving plates **315***a* may be configured to support the left and right lower ends of the battery module **200** in the upward direction. The 2 receiving plates **315***a* may act as a stopper to prevent the battery module **200** disposed at the lower position from moving up.
(52) The 2 receiving plates **315***a* may be spaced apart in the horizontal direction as much as the horizontal width of the battery module **200**. The 2 receiving plates **315***a* may be configured such that the front and rear ends are coupled to the front frame **311** and the rear frame **313**.

(53) When the plurality of receiving plates **315***a* is arranged in the vertical direction, the rack frame **315** may have a coupling bar **315***b* extending in the vertical direction to fix the plurality of receiving plates **315***a*. The coupling bar **315***b* may be configured to be coupled to part of each of the plurality of receiving plates **315***a*. The rack frame **315** may include a fixing bar **315***c* coupled to the upper and lower ends of the coupling bar **315***b* and having the front and rear ends coupled to the front frame **311** and the rear frame **313**.
(54) Referring back to FIGS. **1** to **3**, the front frame **311** may include a plurality of pillars **311***a* extending in the vertical direction. The upper and lower ends of the pillar **311***a* may be coupled to an external object (for example, the bottom or floor, or other rack case **310**). The pillar **311***a* may act as the main framework of the entire hexahedral shape of the rack case **310**. For example, as shown in FIG. **1**, the front frame **311** may include 3 pillars **311***a* disposed at the front end of the rack case **310**.
(55) The front frame **311** may include a connecting part **311***b* connecting the upper or lower end between the pillars **311***a* in the horizontal direction. The connecting part **311***b* may be configured to be connected (coupled) to the upper end or the lower end of the pillar **311***a*. For example, as shown in FIG. **3**, the connecting part **311***b* may be disposed between each of the upper and lower ends of 2 pillars **311***a*.
(56) The rear frame **313** may include a plurality of pillars **313***a* extending in the vertical direction. The upper and lower ends of the pillar **313***a* may be coupled to an external object, for example, the bottom or floor, or other rack case **310**. The pillar **313***a* may act as the main framework of the entire hexahedral shape of the rack case **310**. For example, as shown in FIG. **3**, the rear frame **313** may include 3 pillars **313***a* disposed at the rear end of the rack case **310**.
(57) The rear frame **313** may include a connecting part **313***b* to connect the pillar **313***a* in the horizontal direction. The connecting part **313***b* may be configured to be connected (coupled) to the upper end or the lower end of the pillar **313***a*. For example, as shown in FIG. **3**, the connecting part **313***b* may be disposed between each of the upper and lower ends of 2 pillars **313***a*.
(58) The top plate **317** may be disposed on the uppermost of the plurality of battery modules **200**. The top plate **317** may be configured to be connected to the top of the front frame **311** and the top of the rear frame **313**. The top plate **317** may include an X-shaped beam **317***a* to reinforce the mechanical strength of the top plate **317**. When the top plate **317** is rectangular on the plane, the X-shaped beam **317***a* may be disposed in contact with the corners of the rectangle.
(59) Referring back to FIGS. **1** to **3**, the at least one fixing frame **320** may have a body **322** in the shape of a plate extending along the plurality of battery modules **200**. For example, as shown in FIG. **2**, the 4 fixing frames **320** may be provided on the rear surface of the rear frame **313** of the rack case **310**.
(60) The fixing frame **320** may include a fixing portion **324** formed on one surface of the body **322** and configured to be attached/detached to/from each of the plurality of battery modules **200**. The number of fixing portions **324** corresponding to the plurality of battery modules **200** may be provided. The fixing portion **324** may be configured to be attached/detached to/from the rear cover **260** of the battery module **200**.
(61) That is, the fixing portion **324** may be configured to be installed when the battery rack **300** needs to be delivered. When the battery rack **300** is installed at the installation location after delivery, the fixing portion **324** may be configured to be separated from the plurality of battery modules **200** and the rack case **310**.
(62) According to this configuration of the present disclosure, the battery rack **300** of the present disclosure includes the at least one fixing frame **320** including the fixing portion **324** configured to be attached/detached to/from each of the plurality of battery modules **200**, to prevent deformation of the rack case **310** or failure or damage of the battery module **200** or separation of the battery module **200** from the battery rack **300** due to the movement of the battery rack **300** by the heavy battery module **200** during delivery. Subsequently, when the battery rack **300** is fixed at the

installation location, the fixing frame **320** may be separated from the rack case **310** to deliver the battery rack **300**.
(63) FIG. **5** is a schematic rear perspective view of the battery module of the battery rack according to an embodiment of the present disclosure. FIG. **6** is a schematic rear perspective view of the fixing frame of the battery rack according to an embodiment of the present disclosure. FIG. **7** is a schematic front perspective view of the fixing frame of the battery rack according to an embodiment of the present disclosure.
(64) Referring to FIGS. **5** to **7**, each of the plurality of battery modules **200** may include a hanger member **270** having a part of which is fixed to the outer surface. The hanger member **270** may be provided on the outer side of the rear cover **260** of the battery module **200**. The part of the hanger member **270** may be firmly coupled to the rear cover **260**. The hanger member **270** may be detached from the fixing portion **324**. The hanger member **270** may be configured to be hook-coupled to the fixing portion **324** of the fixing frame **320**.
(65) As shown in FIG. **7**, the fixing frame **320** may include 6 fixing portions **324**. The fixing portion **324** may have a hook structure **324***h* configured to be hook-coupled to the hanger member **270**.
(66) According to this configuration of the present disclosure, the present disclosure is configured such that the hanger member **270** installed in each of the plurality of battery modules **200** and the fixing portion **324** of the fixing frame **320** are hook-coupled to each other, thereby making it easy to couple each of the plurality of battery modules **200** by hanging the fixing frame **320** on the hanger member. After the delivery of the battery rack **300** is completed, when the battery rack **300** is fixed to the installation location, it is possible to easily decouple the fixing frame **320** from each of the plurality of battery modules **200**, thereby performing the operation of removing the fixing frame **320** in a simple and quick manner.
(67) FIG. **8** is a schematic partial plane view of the battery module and the fixing frame according to an embodiment of the present disclosure. FIG. **9** is a schematic enlarged plane view of section A in FIG. **8**.
(68) Referring to FIGS. **8** and **9** together with FIGS. **5** to **7**, the hanger member **270** may include a hanger bar **272** and a leg **274**. The hanger bar **272** may be in the shape of a flat plate in which the hook structure **324***h* is inserted and fixed on one side. The hanger bar **272** may be configured such that the flat plate stands upright in the vertical direction. The hanger bar **272** may be spaced a predetermined distance apart from the outer surface of the battery module **200**.
(69) The leg **274** may extend from each of the two ends of the hanger bar **272**. The leg **274** may have a portion extending in the diagonal direction and a portion connected to the outer surface of the rear cover **260**. The leg **274** may be weld-coupled or bolt-coupled to the outer surface of the battery module **200**.
(70) The leg **274** may have a bent part K**1** that is bent in the diagonal direction such that the hanger bar **272** is spaced a predetermined distance apart from the outer surface of the battery module **200**. That is, the leg **274** may have the bent part K**1** that is bent in the rearward direction from two ends of the leg **274**. The rearwardly bent part K**1** may be configured to be connected to the hanger bar **272**.
(71) According to this configuration of the present disclosure, the hanger member **270** includes the hanger bar **272** spaced a predetermined distance apart from the outer surface of the battery module **200** and the leg **274** having the end extending from the hanger bar **272** fixed to the outer surface the battery module **200**, and thus it is easy to couple or separate the hook structure **324***h* of the fixing portion **324** of the fixing frame **320** to/from the hanger member **270**, thereby performing the installation operation in a simple and quick manner.
(72) Referring back to FIGS. **8** and **9**, the fixing frame **320** may further include a support portion **326** extending from the body **322** such that one surface wraps around at least part of the outer surface of the leg **274**. The support portion **326** may be disposed on the left and right sides of the

hook structure **324***h*. The support portion **326** may extend in the vertical direction along the body **322**. The support portion **326** may extend diagonally in the left and right direction. Due to this structure, the support portion **326** may prevent the hanger member **270** from moving to the left and right. For example, after the battery rack **300** is installed, when an earthquake occurs, the support portion **326** may prevent the battery module **200** from moving to the left and right due to the earthquake.

(73) According to this configuration of the present disclosure, the fixing frame **320** of the present disclosure further includes the support portion **326** extending from the body **322** such that one surface wraps around at least part of the outer surface of the leg **274**, to prevent the battery module **200** from moving to the left and right during delivery. Accordingly, it is possible to prevent deformation of the rack case **310** caused by the leftward and rightward movement of the battery module **200**.

(74) Referring back to FIGS. **6** and **7** together with FIG. **2**, the fixing frame **320** may include a case coupling portion **328**. The case coupling portion **328** may be provided at the upper and lower ends of the body **322**. The case coupling portion **328** may be coupled to the rack case **310**.

(75) In detail, the rack case **310** may include a connecting bar **318** coupled to the case coupling portion **328**. The connecting bar **318** may have two ends coupled to the rear frame **313**. For example, as shown in FIG. **3**, the case coupling portion **328** provided at the upper and lower ends of the body **322** may be configured to be connected to 2 connecting bars **318** spaced apart from each other in the vertical direction. In this instance, the case coupling portion **328** may be bolting-coupled to the connecting bar **318**. As shown in FIG. **3**, a total of 6 connecting bars **318** coupled to the case coupling portion **328** of each of 4 fixing frames **320** may be provided each two connecting bars **318** for each of the upper, intermediate and lower ends of the rack case **310**.

(76) According to this configuration of the present disclosure, the case coupling portion **328** coupled to the rack case **310** is provided at the upper and lower ends of the body **322** of the fixing frame **320**, and thus the fixing frame **320** and the rack case **310** may be fixed with a strong fixing strength, and the fixing frame **320** may strongly hold the plurality of battery modules **200**. It is possible to effectively prevent deformation during delivery of the battery rack **300**. It is possible to effectively increase the durability of the battery rack **300**.

(77) FIG. **10** is a schematic rear perspective view of a fixing frame according to another embodiment of the present disclosure.

(78) Referring to FIG. **10**, the fixing frame **320**A of FIG. **10** may have the same components as the fixing frame **320** of FIG. **6**, except further comprising a wall coupling portion **329**. A detailed description of the above-described components of the fixing frame **320**A is omitted herein.

(79) The wall coupling portion **329** of the fixing frame **320**A may be configured to fix the fixing frame **320**A to a wall W adjacent to the battery rack **300**. The wall coupling portion **329** may have a coupling hole H**4** for coupling with the wall W and a bolt (not shown) that is inserted into the coupling hole H**4**. The wall W may have a separate fixing element (not shown) for coupling with the wall coupling portion **329**. The fixing element may have a fixing hole that is in communication with the coupling hole H**4**.

(80) However, the wall coupling portion **329** is necessarily fixed to the wall W using only a fixing element, and the wall coupling portion **329** may be directly coupled to the wall W having a fixing hole H**3**. The wall coupling portion **329** may be fixed to the wall W through the bolt inserted into the coupling hole H**4** of the wall coupling portion **329** and the fixing hole H**3** of the wall W.

(81) According to this configuration of the present disclosure, the wall coupling portion **329** of the fixing frame **320**A is configured to fix the fixing frame **320**A to the wall W adjacent to the battery rack **300**, to couple and fix the upper part of the rack case **310** to the wall W of the storage location. It is possible to prevent the upper part of the battery rack **300** from moving back and forth and to the left and right during delivery. It is possible to prevent the rack case **310** of the battery rack **300** from being deformed, or the plurality of battery modules **200** from being deformed or damaged by

the rack case **310**.
(82) FIG. **11** is a schematic partial perspective view of the lower part of the battery rack according to an embodiment of the present disclosure.
(83) Referring to FIG. **11** together with FIG. **3**, the front frame **311** and the rear frame **313** may have bent portions **311***c*, **313***c* having one surface parallel to the ground at the lower end respectively. The bent portion **311***c* may have a bent shape to be joined to the bottom where the battery rack **300** is stored.
(84) For example, the bent portion **311***c* of the front frame **311** may not be separated from the pillar **311***a*, and may be integrally formed with the pillar **311***a*. That is, one surface of the lower end of the pillar **311***a* may be bent in contact with the bottom to form the bent portion **311***c*.
(85) The bent portion **313***c* of the rear frame **313** may not be separated from the pillar **313***a*, and may be integrally formed with the pillar **313***a*. That is, one surface of the lower end of the pillar **313***a* may be bent in contact with the bottom to form the bent portion **313***c*.
(86) The bent portion **311***c* of the front frame **311** may have a bolting hole H**1**. The bolting hole H**1** is configured to be coupled to the bottom where the battery rack **300** is stored, and a fixing bolt (not shown) may be inserted into the bolting hole H**1** for coupling with the bottom. Alternatively, the bolting hole H**1** may be configured for coupling with another rack case **310**.
(87) The bent portion **313***c* of the rear frame **313** may have a slit **313***c***1** into which the body of a bolt B**1** is slidably inserted in the horizontal direction S. That is, since the rear frame **313** may be disposed in contact with or close to the wall of the installation space of the battery rack **300** or another battery rack, an operator cannot have the operational space for performing the bolting operation at the rear of the rear frame **313** to fix the battery rack **300** to the bottom.
(88) The inventor pre-installed the bolt B**1** in the bottom to a predetermined depth such that the bolt B**1** stands upright vertically from the ground, and slidably moved back the battery rack **300** to insert the body of the bolt B**1** into the slit **313***c***1**. Subsequently, the operator may tighten the bolt B**1** inserted into the slit **313***c***1** using a long rod-shaped spanner in front of the battery rack **300** to fix the rear frame **313** to the bottom.
(89) According to this configuration of the present disclosure, the front frame **311** and the rear frame **313** include the bent portions **311***c*, **313***c* having one surface parallel to the ground at the lower end thereof and the bolting hole H**1** and the slit **313***c***1** formed in the bent portions **311***c*, **313***c* respectively, to fix the battery rack **300** to the bottom of the storage location with a strong fixing force, thereby effectively preventing the battery rack **300** from tilting or moving during delivery. It is easy to fix the battery rack **300** to the installation location.
(90) FIG. **12** is a schematic partial perspective view of the upper part of the battery rack according to an embodiment of the present disclosure. FIG. **13** is a schematic partial perspective view of a battery rack according to another embodiment of the present disclosure.
(91) Referring to FIGS. **12** and **13** together with FIG. **3**, the battery rack **300** may include two or more rack cases **310** stacked in the vertical direction. In detail, among the two or more rack cases **310**, the lower part of the pillars **311***a*, **313***a* of the rack case **310** disposed at the upper position and the upper end of each of the pillars **311***a*, **313***a* of the rack case **310** disposed at the lower position may be coupled to each other.
(92) In further detail, a bolting hole H**2** configured to be coupled to another rack case **310** may be provided in the upper end of the pillar **311***a* of each of the front frame **311** and the rear frame **313**. That is, the bent portion **311***c* provided below the pillar **311***a* of the rack case **310** disposed at the upper position and the bent portion **311***c* provided on the pillar **311***a* of the rack case **310** disposed at the lower position may be coupled to each other by a bolt (not shown) passing through the bolting hole H**2** and the bolting hole H**1**.
(93) For example, as shown in FIGS. **11** and **12**, the bolting hole H**2** in communication with the bolting hole H**1** of the bent portion **311***c* provided below the pillar **311***a* of the rack case **310** disposed at the upper position may be provided in the upper end of the pillar **311***a* of the rack case

310 disposed at the lower position. That is, the fixing bolt may be inserted into the bolting hole H**1** of the bent portion **311***c* of the rack case **310** disposed at the upper position and the bolting hole H**2** provided in the upper end of the pillar **311***a* of the rack case **310** disposed at the lower position.
(94) FIG. **14** is a schematic partial horizontal cross-sectional view of the battery rack of FIG. **1** taken along the line A-A′.
(95) Referring to FIGS. **1** and **14**, the pillars **311***a*, **313***a* of each of the front frame **311** and the rear frame **313** of the present disclosure may include an outer pillar **311***a***1** extending in the vertical direction and having one open horizontal side and an inner pillar **311***a***2** inserted into the outer pillar **311***a***1** and extending in the vertical direction.
(96) In detail, the outer pillar **311***a***1** may be a C shaped steel that is open to one side when viewed in the horizontal cross section. The inner pillar **311***a***2** may be in the shape of a rectangular pipe. The outer pillar **311***a***1** and the inner pillar **311***a***2** may include different materials. The inner pillar **311***a***2** may be disposed in the outer pillar **311***a***1** to support the outer pillar **311***a***1** against the horizontal outer side.
(97) According to this configuration of the present disclosure, the pillar **311***a* includes the outer pillar **311***a***1** extending in the vertical direction and having one open horizontal side and the inner pillar **311***a***2** inserted into the outer pillar **311***a***1** and extending in the vertical direction, thereby effectively increasing the mechanical strength of the rack case **310**. That is, instead of the single pillar, the combination of the outer pillar **311***a***1** and the inner pillar **311***a***2** may ensure a higher mechanical strength than the single body configuration without greatly increasing the weight of the rack case **310**.
(98) FIGS. **15** and **16** are schematic rear views of the fixing frame of the battery rack according to another embodiment of the present disclosure.
(99) Referring to FIGS. **15** and **16** together with FIGS. **2** and **7**, the case coupling portion **328** disposed below the fixing frame **320** may have 2 bolting holes H**5**, H**6**. The 2 bolting holes H**5**, H**6** may be spaced a predetermined distance apart from each other in the vertical direction. For example, as shown in FIG. **15**, when the case coupling portion **328** is bolting-coupled to the connecting bar **318** through the upper bolting hole H**5**, the hook structure **324***h* of the fixing portion **324** may be hook-coupled to the hanger bar **272** of the hanger member **270** by the downward movement of the fixing frame **320**.
(100) On the contrary, for example, as shown in FIG. **16**, when the case coupling portion **328** is bolting-coupled to the connecting bar **318** through the lower bolting hole H**6**, compared to the coupling of the upper bolting hole H**5** as previously described, the fixing frame **320** may move up and be coupled to the connecting bar **318**, and the hook structure **324***h* of the fixing portion **324** may be separated from the hanger bar **272** of the hanger member **270**.
(101) Referring to FIG. **6**, the case coupling portion **328** disposed on the fixing frame **320** may have a bolting hole H**7** extending in the vertical direction. In this instance, the extended length of the bolting hole H**7** may be similar to or longer than the distance between the upper bolting hole H**5** and the lower bolting hole H**6**.
(102) That is, the upper bolting hole H**5** or the lower bolting hole H**6** of the fixing frame **320** may be coupled to the connecting bar **318**, and accordingly the upper and lower positions of the bolt coupled to the bolting hole H**7** of the case coupling portion **328** disposed on the fixing frame **320** may be changed.
(103) In detail, when the case coupling portion **328** is bolting-coupled to the connecting bar **318** through the upper bolting hole H**5**, the bolt inserted into the bolting hole H**7** may be disposed on the bolting hole H**7**. On the contrary, when the case coupling portion **328** is bolting-coupled to the connecting bar **318** through the lower bolting hole H**6**, the bolt inserted into the bolting hole H**7** may be disposed lower than the bolting hole H**7**.
(104) That is, the previous example describes an embodiment in which the fixing frame **320** is removed after the delivery of the battery rack **300**. In contrast, in another embodiment of the

present disclosure, the case coupling portion **328** may be bolting-coupled to the connecting bar **318** through the lower bolting hole H**6** to fix the hook structure **324***h* of the fixing portion **324** separated from the hanger bar **272** of the hanger member **270** to the rack case **310** without removing the fixing frame **320**.

(105) After the battery rack **300** is installed in the installation location, when a defective or old battery module **200** needs replacement, the battery module **200** mounted in the rack frame **315** is moved forward and removed, and a new battery module **200** is moved back from the rack frame **315** and is mounted. In this instance, the fixing portion **324** may move the battery module **200** back and forth when the hook structure **324***h* is kept in separated state from the hanger bar **272** of the hanger member **270**. Even when the hook structure **324***h* is separated from the hanger bar **272** of the hanger member **270**, since the support portion **326** extending in the vertical direction from the body **322** wraps at least part of the outer surface of the leg **274** of the hanger member **270**, the fixing frame **320** may keep preventing the battery module **200** from moving to the left and right.

(106) Accordingly, even when the battery rack **300** is installed in a location at which earthquakes are likely to occur, the fixing frame **320** may prevent the mounted battery module **200** from moving to the left and right due to the earthquake. Accordingly, it is possible to greatly improve the earthquake-resistant stability of the battery rack **300**.

(107) According to this configuration of the present disclosure, the fixing frame **320** has the 2 bolting holes H**5**, H**6** spaced apart from each other in the vertical direction in the case coupling portion **328**, to ease the operation of replacing the battery module **200** provided in the battery rack **300**, and prevent the battery module mounted in the rack case **310** from moving to the left and right, thereby effectively improving the earthquake resistance and durability of the battery rack **300**.

(108) Referring to FIGS. **15** and **16**, the fixing frame **320** of the present disclosure may further include an indicator. The indicator may be configured to indicate whether or not the hook structure **324***h* of the fixing portion is coupled to the hanger bar **272** of the hanger member **270** in color. The indicator may have stickers **321***a*, **321***b* of different colors. For example, the indicator may have a red sticker **321***a* and a green sticker **321***b*.

(109) For example, as shown in FIG. **16**, the red sticker **321***a* and the green sticker **321***b* may be attached to the outer surface of the connecting bar **318**. Accordingly, for the fixing frame **320** to be coupled to the hanger member **270** of the plurality of battery modules **200**, when the fixing frame **320** is coupled to the connecting bar **318** using a bolt inserted into the upper bolting hole H**5**, part of the red sticker **321***a* may be exposed to the outside through the lower bolting hole H**6**.

(110) As shown in FIG. **15**, to decouple the fixing frame **320** and the hanger member **270** of the plurality of battery modules **200**, when the fixing frame **320** is coupled to the connecting bar **318** by inserting a bolt into the lower bolting hole H**6**, part of the green sticker **321***b* may be exposed to the outside through the upper bolting hole H**5**.

(111) According to this configuration of the present disclosure, the present disclosure includes the indicator to allow a user to easily see the coupling of the fixing frame **320** and the hanger member **270** with the eye, thereby managing the battery rack fast, and easing the operation of separating a failed battery module **200** among the plurality of battery modules **200** from the rack case **310**.

(112) Although not shown, the battery rack **300** according to the present disclosure may further include, for example, a Battery Management System (BMS) (not shown), inside or outside of the rack case **310**.

(113) A power storage system according to the present disclosure may include at least one battery rack **300** according to the present disclosure. In particular, the power storage system may include a plurality of battery racks **300** according to the present disclosure. The plurality of battery rack **300** may be electrically connected to each other. The power storage system according to the present disclosure may be implemented in various forms, for example, smart grid systems or electricity charging stations.

(114) The terms indicating directions as used herein such as upper, lower, left, right, front and rear are used for convenience of description only, and it is obvious to those skilled in the art that the term may change depending on the position of the stated element or an observer.
(115) While the present disclosure has been hereinabove described with regard to a limited number of embodiments and drawings, the present disclosure is not limited thereto and it is obvious to those skilled in the art that various modifications and changes may be made thereto within the technical aspects of the present disclosure and the equivalent scope of the appended claims.
DESCRIPTION OF REFERENCE NUMERALS
(116) TABLE-US-00001 300: Battery rack 100: Cell assembly 200: Battery module 210: Module housing 110: Secondary battery 320: Fixing frame 310: Rack case 270: Hanger member 322, 324: Body, Fixing portion 324h: Hook structure 272, 274: Hanger bar, Leg 328: Case coupling portion 326: Support portion 311b, 313b: Connecting part 329: Wall coupling portion H1, H2: Bolting hole 321, 321a, 321b: Indicator, Red sticker, 224, 226: Top, Side Green sticker 260: Rear cover 311, 313, 315, 317: Front frame, Rear frame, Rack frame, Top plate 311a, 313a: Pillar 311c: Bent portion 220: Upper cover 240: Base plate H5, H6: Upper bolting hole, Lower bolting hole 250: Front cover 311a1, 311a2: Outer pillar, Inner pillar 321a, 321b: Stickers
INDUSTRIAL APPLICABILITY
(117) The present disclosure relates to a battery rack. In addition, the present disclosure can be used in the industry related to an energy storage system comprising the battery rack.

Claims

1. A battery rack comprising: a plurality of battery modules arranged in a first direction; at least one rack case including a plurality of rack frames configured to mount the plurality of battery modules, the at least one rack case having two side walls, a top and a bottom; at least one fixing frame connected to the at least one rack case, the at least one fixing frame including a body in the shape of a plate extending along the plurality of battery modules; and a fixing portion formed on one surface of the body and configured to be attached to each of the plurality of battery modules or detached from each of the plurality of battery modules; a plurality of hanger members respectively coupled to an outer surface of the plurality of battery modules, wherein the fixing portion of the fixing frame has a plurality of hook structures, wherein each hanger member of the plurality of hanger members is configured to be coupled to a respective one of the plurality of hook structures of the fixing portion of the fixing frame, and wherein each hanger member of the plurality of hanger members includes: a hanger bar in the shape of a plate in which the hook structure is inserted and fixed on one side, the hanger bar spaced a predetermined distance from the outer surface of the battery module; and a leg extending from each of two ends of the hanger bar, wherein each leg is fixed to the outer surface of the battery module, and wherein each leg has a bent portion.
2. The battery rack according to claim 1, wherein each of the plurality of battery modules includes a module housing including an upper cover, a base plate, a front cover and a rear cover, and wherein each hanger member of the plurality of hanger members is disposed on an outer side of the rear cover of a respective one of the plurality of module housings.
3. The battery rack according to claim 1, wherein the at least one fixing frame includes a support portion extending vertically along the body and extending from the body such that a surface wraps around at least part of an outer surface of each leg.
4. The battery rack according to claim 3, wherein the support portion is configured to wrap around at least part of the outer surface of each leg of the hanger member even when the hook structure is separated from the hanger bar of the hanger member.
5. The battery rack according to claim 1, wherein upper and lower ends of the body of the at least one fixing frame have a case coupling portion that is coupled to the rack case.

6. The battery rack according to claim 5, wherein the rack case includes a connecting bar connected to the case coupling portion to couple the case coupling portion to the rack case.
7. The battery rack according to claim 6, wherein the case coupling portion of the at least one fixing frame includes: an upper bolting hole configured to couple the plurality of hook structures to the plurality of hanger members when coupled to the connecting bar; and a lower bolting hole spaced from the upper bolting hole at a predetermined distance in the vertical direction and configured to separate the plurality of hook structures from the plurality of hanger members when coupled to the connecting bar.
8. The battery rack according to claim 7, wherein the fixing frame further includes an indicator configured to indicate whether the plurality of hook structures is coupled to the plurality of hanger members.
9. The battery rack according to claim 1, wherein the fixing frame further includes a wall coupling portion configured to fix the fixing frame to a wall adjacent to the battery rack.
10. The battery rack according to claim 1, wherein the rack case includes: a front frame including a plurality of pillars disposed at a front end of the plurality of battery modules and extending in a vertical direction, and a connecting part connecting the pillars in a horizontal direction; a rear frame including a plurality of pillars disposed at a rear end of the plurality of battery modules and extending in the vertical direction and a connecting part connecting the pillars in the horizontal direction; and wherein a lower end of each of the front frame and the rear frame has a bent portion having one surface parallel to the ground, and a bolting hole formed in the bent portion.
11. The battery rack according to claim 10, wherein the battery rack includes two or more rack cases stacked in a vertical direction, wherein each pillar of the front frame and the rear frame has a bolting hole configured to be coupled to another rack case at an upper end, wherein a bolting hole is disposed at the bent portion of the pillar, and wherein the rack case disposed at an upper position is bolt-coupled to a bolting hole of the pillar of the rack case disposed at a lower position.
12. The battery rack according to claim 10, wherein each pillar includes an outer pillar extending in a vertical direction and having one open horizontal side and an inner pillar inserted into the outer pillar and extending in the vertical direction.
13. The battery rack according to claim 12, wherein the outer pillar is a C-shaped steel that is open to one side when viewed in a horizontal cross section, and the inner pillar is a rectangular pipe.
14. The battery rack according to claim 10, wherein the bent portion of the rear frame has a slit into which a body of a bolt is slidably inserted in a horizontal direction.
15. An energy storage system comprising at least one battery rack according to claim 1.
16. The battery rack according to claim 1, wherein the at least one fixing frame extends in the first direction.
17. The battery rack according to claim 1, wherein the at least one fixing frame is attached to a rear of the at least one rack case between side edges of the at least one rack case.
18. The battery rack according to claim 1, wherein the plurality of hook structures extend toward the plurality of battery modules.